

Lecture 7: Greedy Algorithms

Professor Υ 's Greedy Algorithms



Attempt 1: Choose _____,
take it, remove overlaps, iterate on remaining intervals.

Untangling String Theory

Poetry for Physicists

REM Lab

Aliens

Culture of Hieroglyphs

Foundations of CS

Blah

Applied Topology with Doughnuts

Advanced Blah

Counterexample:

What makes
this a greedy
algorithm?



Professor Υ 's Greedy Algorithms



Attempt 2: Choose _____,
take it, remove overlaps, iterate on remaining intervals.

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Counterexample:

Professor Υ 's Greedy Algorithms



Attempt 3: Choose _____,
take it, remove overlaps, iterate on remaining intervals.

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Counterexample:

Task Selection Correctness Proof

Key Claim: An interval **I** that _____
is a **safe** choice, i.e., it is in some optimal solution.

Proof: Consider any optimal solution **OPT**. Let **J** be the first interval in **OPT**.

- **J** ends no earlier than **I** because _____
- For every other interval **J'** in **OPT**: _____
so $\text{end}(\mathbf{I}) \leq \text{start}(\mathbf{J}')$.
- Thus, we can **exchange J for I** in **OPT** and get a valid solution **OPT'** of the same (optimal) size!

Proof that **all** chosen intervals are “safe”

Let **ALG** be the output of the EET algorithm. If **ALG**=**OPT**, we’re done.

If **ALG**≠**OPT**, consider the **first** interval I_{diff} of **ALG** that’s **not** in **OPT**.

$I_1, I_2, I_3, \dots, I_{\text{diff}}, \dots = \text{ALG}$ (ordered by end time)

$\underbrace{I_1, I_2, I_3, \dots}_{\text{same as ALG}}, \underbrace{J_{\text{diff}}, \dots}_{\text{different from ALG}} = \text{OPT}$ (ordered by end time)

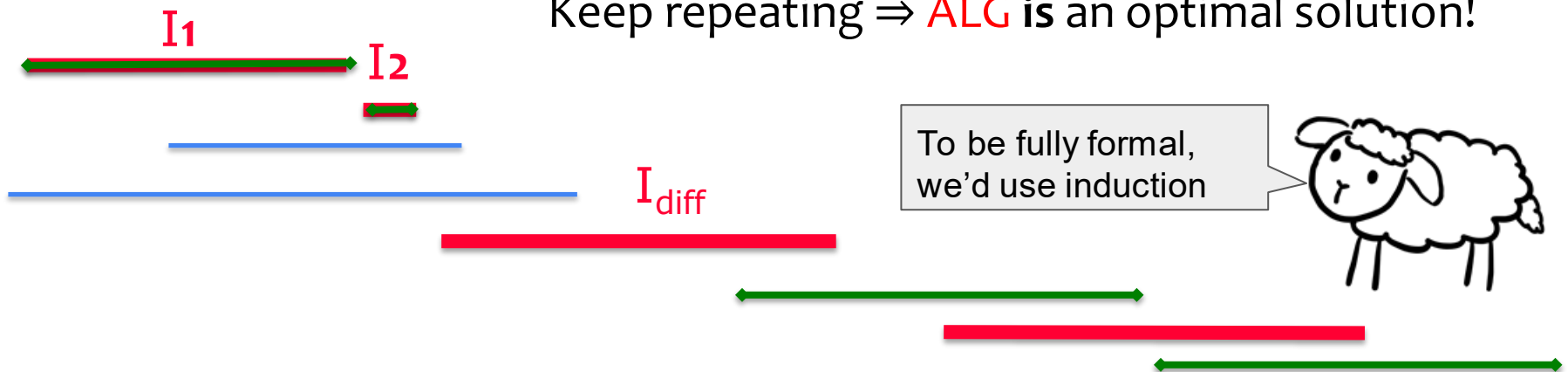
(will prove)

Key Claim: There’s an optimal solution **OPT’** containing $I_1, I_2, I_3, \dots, I_{\text{diff}}$.

Consequence of Claim: We moved the “first difference” point later!

(I.e., first difference of **ALG** ↔ **OPT’** is after that of **ALG** ↔ **OPT**)

Keep repeating $\Rightarrow$ **ALG** is an optimal solution!



Proof that **all** chosen intervals are “safe”

Key Claim: There's an optimal solution OPT' containing $I_1, I_2, I_3, \dots, I_{\text{diff}}$.

Proof of Claim: Let _____ be _____

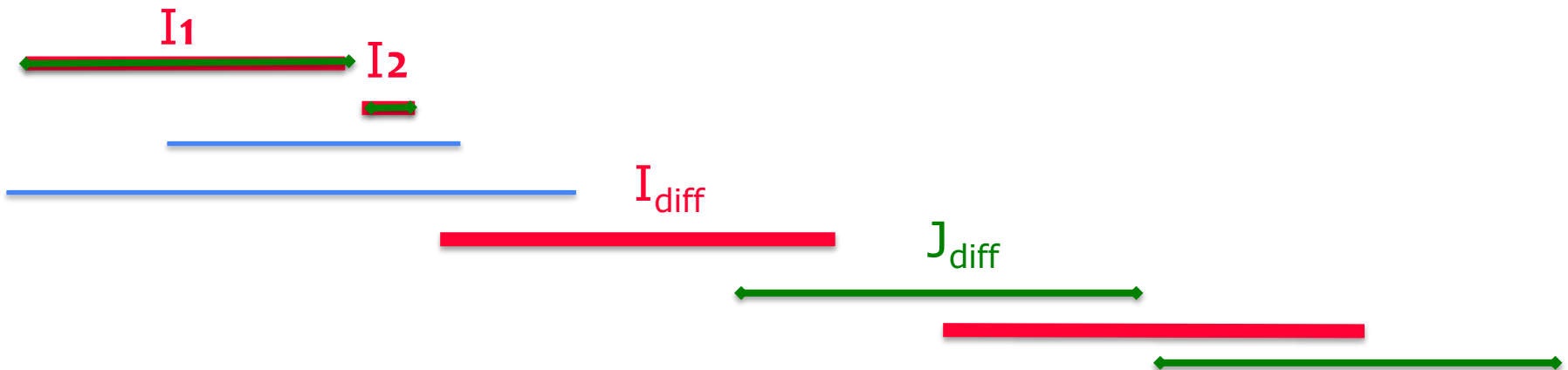
Take OPT and **exchange** _____ out for I_{diff} to form OPT' .

→ Why is OPT' of optimal size?

→ Why is OPT' a valid solution? Why doesn't I_{diff} overlap ...

◆ with $I_1, I_2, I_3, \dots, I_{\text{diff}-1}$ in OPT' ?

◆ with $J_{\text{diff}+1}, J_{\text{diff}+2}, \dots$ in OPT' ?

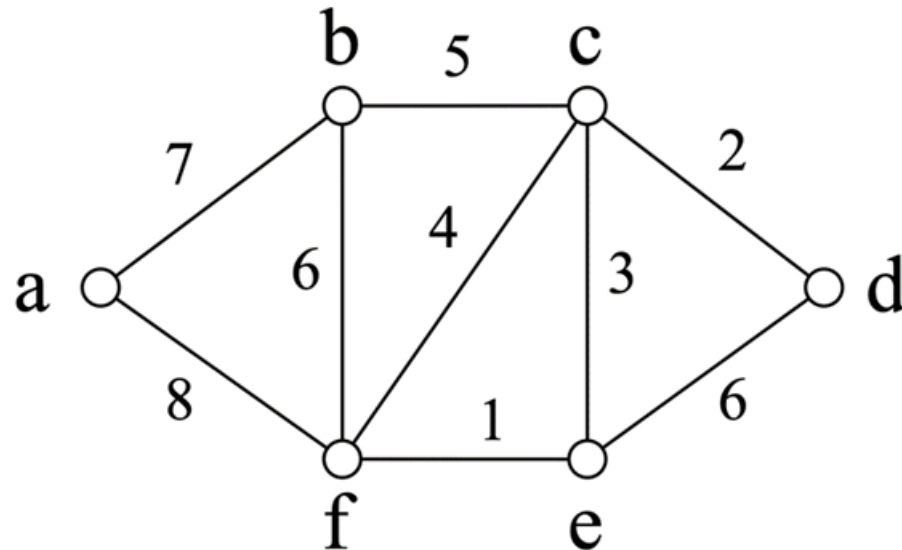


Another problem where greed is good...

A Highway Problem

Claim: Solution is _____. This fact will be useful later too
Why?

Definition (review): A _____ is _____.



Kruskal's Algorithm: Correctness (continued):

$OPT \cup \{e_{diff}\}$ must have _____, call it C

_____, so there must be at least one edge in C
_____, call it f

Form OPT' by taking OPT and exchanging e_{diff} for f

After the exchange, why is the graph still a tree?

Still connected? Still no cycles?

Why did the exchange not increase the weight of the spanning tree?

Want to show: $\text{weight}(e_{diff}) \leq \text{weight}(f)$.